UMANG VEKARIYA

AI/ML | Full-Stack Development | Software Engineering

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EDUCATION

Vellore Institute of Technology (VIT), Vellore Expected May 2028
B.Tech in Computer Science (AI & ML) | **CGPA: 9.21/10** | VITEEE Rank: 29

Class XII - FIITJEE Junior College
92.5% | 2022-2024

Class X - Narayana School
GPA: 9.6/10 | 2021-2022

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, TypeScript
AI/ML & Vision: scikit-learn, TensorFlow, Ollama, OpenCV, MediaPipe, NumPy, SciPy
Web & Data: React, FastAPI, REST APIs, SQLAlchemy, PostgreSQL, MySQL, SQLite
Cloud & Tools: Azure VMs, Blob Storage, Queue Storage, Docker, Git

EXPERIENCE & TRAINING

Software Engineering Intern | Mahant Solutions May-Jul 2026
- Supported full-stack development, debugging, testing and application maintenance.

ACM India Summer School | IISc Bengaluru Jul 2026
Selected participant in **Edge AI & Robotics**; hands-on training in embedded ML, computer vision and robotics.

SELECTED PROJECTS

Intent Lab | Python, scikit-learn, Ollama Case study
- Built a local **Qwen3-4B** planner with a TF-IDF/logistic-regression action checker, explicit permission gates and execution receipts.
- Grouped train/validation/test splits by request lineage and froze model selection; completed **11/12 synthetic mock tasks versus 7/12** for the prior agent/checker workflow.

RealityForge | TypeScript, MediaPipe, WebAssembly Case study
- Built a browser simulation that converts calibrated desk silhouettes into obstacles, with MediaPipe hand controls and gesture classification.
- Implemented **Hungarian-assignment tracking** and occlusion-aware ID retention; checked **6 synthetic clips / 144 frames** and passed **27 perception tests**.

PlayableReplay | Python, OpenCV, NumPy, JavaScript Case study
- Reconstructed fixed-camera Breakout footage into playable offline browser games using pixel tracking, collision extraction, physics fitting and deterministic replay.
- Fitted on the first **60% of observations** across **5 synthetic fixtures**; refinement reduced mean error from **46.8 to 7.63 px** on one Classic fixture (later 40%: **15.87 px**).

VIT Vault | FastAPI, PostgreSQL, Azure, Docker Case study
- Co-developed an Azure platform for versioned submissions, private feedback and Docker labs; added durable LAN receipts and queued cloud verification.
- Verified controlled upstream-failure and restart recovery against Azure, retaining receipts, preventing duplicate cloud submissions and checking downloaded files with **SHA-256**.

PolyGlot | Python, JavaScript, Pyodide/WebAssembly Case study
- Built an educational compiler for **6 restricted syntax families**, with type checking, optimized IR, validated bytecode artifacts and a bounded VM.
- Shared one Python compiler across CLI and browser workers; passed **298 compiler/API tests and 30 frontend helper checks**.

CERTIFICATIONS

Meta Front-End Developer Professional Certificate | Oracle Cloud Infrastructure AI Foundations